llmXive Automated Review of Claw-SWE-Bench: A Benchmark for Evaluating OpenClaw-style Agent Harnesses on Coding Tasks

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper presents a rigorous benchmark for evaluating agent harnesses, with most quantitative claims supported by the provided tables. However, a critical numerical inconsistency exists in Table 1 (Section 5.1) regarding the "Apply Failed" metric for the bare adapter. The table lists 67 resolved instances (19.1% of 350) but reports an "Apply Failed" rate of 69.1%. Mathematically, if 67 instances were resolved, 283 failed, which corresponds to an 80.9% failure rate, not 69.1%. This discrepancy suggests either a calculation error in the percentage or a misalignment between the "Resolved" count and the "Apply Failed" definition (e.g., if "Apply Failed" only counts non-resolved instances that attempted a patch). This must be corrected to ensure the accuracy of the diagnostic results.

Additionally, the claim in Section 5.2 that the harness spread is 12.5 pp on GLM 5.1 is derived from the difference between OpenClaw (73.4%) and Nanobot (60.9%). While this calculation is correct, the text implies a general "spread" without explicitly defining it as the range between the best and worst performers. Given that GenericAgent (63.1%) is also listed, clarifying that the 12.5 pp figure represents the full range (max - min) would prevent misinterpretation of the variance.

Finally, the repository coverage claim of 79% (34 of 43) in Section 1 and the abstract should be double-checked against the actual subset composition to ensure the percentage is precise and not a rounded approximation that obscures the exact count. These issues are primarily factual corrections and clarifications rather than fundamental flaws in the research design.

Action items.

- [**science**] Table 1 reports 'Apply Failed' as 69.1% for the bare adapter, but 67/350 resolved implies ~80.9% failure. Reconcile the 'Resolved' count with the 'Apply Failed' percentage or clarify the metric definition.
- [**writing**] Section 5.2 claims a 12.5 pp harness spread on GLM 5.1 (OpenClaw vs. Nanobot). Explicitly state this is the range between best and worst to avoid ambiguity, as GenericAgent (63.1%) is closer to the mean.
- [**writing**] Verify the repository coverage claim (34/43 = 79%) in Section 1 and Appendix D.0. Ensure the 79% figure is mathematically precise and consistent with the actual subset composition.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens *runs on a vision-capable model.*

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The diagram effectively visualizes the adapter architecture and the contract mismatch between harnesses and SWE-bench. However, the caption contains a filename reference (‘[C_figure3.png]’) that contradicts the figure label and content, suggesting a copy-paste error.

Figure 2

The figure is incomplete; the caption describes three sub-panels (a, b, c), but the image only displays the scatter plot for panel (a). Additionally, the caption text contains a formatting artifact regarding the model count.

Figure 3

Figure 3 effectively visualizes the impact of future-commit cleanup on Pass@1 scores for nine models. The dumbbell plot clearly shows performance drops for all models, with the legend, axis labels, and right-hand ‘Change’ column providing complete context that matches the caption.

Action items.

- [**writing**] Figure 1: The caption filename ‘[C_figure3.png]’ contradicts the figure label ‘Figure 1’ and the content (which describes the adapter architecture, not the ‘future-commit cleanup’ described in the actual Figure 3 caption).
- [**science**] Figure 2: The caption describes three sub-panels (a, b, c), but the rendered image contains only a single scatter plot corresponding to panel (a). Panels (b) and (c) are missing.
- [**writing**] Figure 2: The caption references ‘5 claws

  $$2sharedmodels'forpanel(b), butthetextcontainsaformattingartifact('$$

  ‘) and the panel itself is not visible.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on domain-specific shorthand and undefined acronyms that hinder accessibility for non-specialist readers. The most critical issue is the use of “claw” as a generic term for “agent harness” (e.g., “five-claw,” “claw sweep”) without a clear definition linking it to the broader concept of an agent framework. This appears to be internal jargon from the “OpenClaw” project that has been exported to the general text.

Additionally, the acronym “pp” (percentage points) is used extensively (e.g., “29.4 pp,” “12.5 pp”) in the Abstract, Introduction, and Results without being defined at first use. While common in statistics, it should be explicitly defined for a general audience.

The term “K-sweep” and the notation “K*” (Section 3, Figure 3) are used to describe a sensitivity analysis of subset size. These are opaque to readers outside the specific optimization context of the authors. Replacing these with “subset-size sensitivity analysis” and “optimal subset size” would improve clarity.

Finally, “L1 difference” (Section 3.2) and “cache hit rate” (Section 5) are used without immediate definition or formulaic context in the main text, relying on the reader to infer the specific mathematical or operational meaning. Defining these terms upon first appearance is necessary to meet the standard of inclusive technical writing.

Action items.

- [**writing**] The manuscript relies heavily on domain-specific shorthand and undefined acronyms that hinder accessibility for non-specialist readers. The most critical issue is the use of “claw” as a generic term for “agent harness” (e.g., “five-claw,” “claw sweep”) without a clear definition linking it to the broader concept of an agent framework. This appears to be internal jargon from the “OpenClaw” project that has been exported to the general text. Additionally, the acronym “pp” (percentage points) is us

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a logically coherent argument for the necessity of a standardized benchmark to isolate the agent harness variable. The central premise—that current evaluations conflate model, harness, and task—is well-supported by the literature review and the proposed experimental design. The introduction of the “adapter protocol” logically follows from the identified “contract mismatch” between OpenClaw-style agents and SWE-bench scoring requirements.

However, there are minor logical gaps in the causal attribution of specific performance metrics:

1. **Confounding of Model and Harness Effects:** The abstract and Section 5.2 claim that “model choice changes Pass@1 by 29.4 pp.” This figure is calculated from the difference between the best and worst models *within the OpenClaw harness* (Table 1). The text presents this as a general property of model choice, but the data does not support a general claim because the harness was not varied for these specific comparisons. The conclusion that model choice is a major factor is valid, but the specific magnitude (29.4 pp) is conditional on

the OpenClaw harness. The logic would be tighter if the claim were qualified (e.g., "Under the OpenClaw harness, model choice...").

2. **Interaction vs. Main Effect:** The paper concludes that "harness choice is a first-order factor" based on the variance observed in the 5-claw x 2-model sweep (Section 5.3). While the variance is indeed large (up to 27.4 pp), the paper explicitly acknowledges in the Conclusion that it "cannot fully decompose harness x model interactions." Logically, if the harness effect is highly dependent on the model (as the 12.5 pp vs. 27.4 pp difference suggests), then "harness choice" is not a standalone first-order factor but an *interaction* term. The conclusion should be refined to state that the *interaction* between harness and model is a first-order factor, or that the harness effect is non-separable from the model choice.
3. **Causal Attribution of Adapter Failure:** The comparison between the "Bare adapter" (69.1% apply failure) and "Full adapter" (<1.5%) in Section 5.1 is used to validate the adapter protocol. However, the "Bare adapter" uses a different prompting strategy (direct diff output) than the "Full adapter" (file edits + Git extraction). The logical leap that the *adapter protocol* alone caused the reduction in failure rate is slightly weakened by the concurrent change in the agent's output modality. It is possible the model simply fails to generate valid diffs when asked directly, regardless of the harness. The paper should clarify that the "Full adapter" success is due to the combination of the protocol *and* the file-edit prompting strategy, rather than the protocol alone.

These issues do not invalidate the paper's core contribution but require slight refinements in the phrasing of causal claims to ensure the conclusions strictly follow from the presented data.

Action items.

- [**writing**] Confounding of Model and Harness Effects: The abstract and Section 5.2 claim that "model choice changes Pass@1 by 29.4 pp." This figure is calculated from the difference between the best and worst models *within the OpenClaw harness* (Table 1). The text presents this as a general property of model choice, but the data does not support a general claim because the harness was not varied for these specific comparisons. The conclusion that model choice is a major factor is valid, but the specific ma
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Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several strong claims regarding the "first-order" nature of harness choice and the generalizability of the benchmark to "OpenClaw-style" agents. While the data supports the specific comparisons made (5 claws, 2 models), the extrapolation to a general class of agents is slightly overreaching.

Specifically, the claim in the Abstract and Introduction that the benchmark covers "8 languages and 43 repositories" is technically accurate for the full set but risks misleading readers about the Lite subset, which the authors admit covers only 34 of those repositories (Section 3.1). The paper should explicitly qualify that the "multilingual" and "43-repo" scope applies to the full benchmark, while the Lite version is a reduced proxy, to avoid overclaiming the representativeness of the Lite set for all 43 repos.

Furthermore, the conclusion that harness choice is a "first-order factor" is derived from a specific set of 5 harnesses with distinct architectural constraints (e.g., stateful vs. stateless, tool deny-lists). The paper does not sufficiently disentangle whether the performance variance is due to the "harness" concept generally or the specific implementation details (like the tool deny-list in `openclaw` vs. the file-based I/O in `generic`). Claiming this applies broadly to "OpenClaw-style" agents without acknowledging that the specific tool configurations might be the primary driver is an overreach.

Finally, the dramatic improvement from 19.1% to 73.4% Pass@1 when switching from the "Bare adapter" to the "Full adapter" (Section 5.1) is framed as the adapter making the agent "scorable." However, the "Bare adapter" failure rate (69.1% apply failed) suggests the issue was primarily a format mismatch (parsing diffs from logs) rather than a lack of problem-solving capability. Framing this as a harness performance gain rather than a protocol correction slightly overstates the adapter's contribution to the agent's intelligence.

Action items.

- **[writing]** The abstract and Section 1 claim the benchmark covers '8 languages' and '43 repositories', but Section 3.1 explicitly states the Lite subset covers only '34 of 43 repositories (79%)'. The paper must clarify if the '43 repositories' claim applies strictly to the full benchmark or if the Lite subset's reduced coverage invalidates the generalizability of the 'multilingual' claim for the Lite version without qualification.
- **[science]** The conclusion states that harness choice is a 'first-order factor' based on a 27.4pp spread on Qwen 3.6-flash. However, the study only evaluates 5 specific harnesses and 2 models. The paper overreaches by implying this finding generalizes to 'OpenClaw-style' agents broadly without acknowledging that the specific tool-deny-list configurations (e.g., in openclaw vs. generic) might be the driver rather than the harness architecture itself.
- **[writing]** Section 5.2 claims the adapter 'makes a general agent scorable' by raising Pass@1

from 19.1% to 73.4%. This conflates the adapter's ability to extract a valid patch with the agent's ability to solve the task. The 19.1% 'Bare adapter' baseline likely fails due to format mismatch, not lack of reasoning. The paper should avoid framing this as a performance gain of the harness/adapter combo over the model's raw capability.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The paper addresses safety and ethics primarily in the "License and Ethics" section (Appendix) and through the design of the execution environment. The authors correctly identify the dual-use nature of autonomous coding agents, noting that stronger agents could be repurposed for "autonomous exploitation." However, the mitigation strategy described is limited to releasing only protocol and instance IDs rather than vulnerable patches. This is a reasonable first step, but the review requires more specificity regarding the content of the benchmark instances themselves.

Specifically, the paper states the workload is drawn from real GitHub issues (SWE-bench-Multilingual and SWE-bench-Verified-Mini). It is unclear whether these issues include known security vulnerabilities (CVEs) or if they are purely functional bugs. If the benchmark includes security-related tasks, the authors must clarify whether they have sanitized the descriptions to prevent the benchmark from serving as a direct "attack guide" for known vulnerabilities. Furthermore, the paper mentions a `REPO_LICENSES.md` file to handle the diverse licenses of the underlying repositories (BSD, Apache, GPL-2). Given that the agents generate code patches, there is a risk of license contamination or violation. The authors should explicitly state whether their evaluation pipeline checks for license compatibility in generated patches or if they place the burden of compliance entirely on the end-user.

Regarding the execution environment, the paper notes that tasks run in SWE-bench Docker images with a 3600-second timeout. While "future-commit cleanup" is mentioned to prevent data leakage from the repository history, there is no explicit mention of network isolation. Autonomous agents attempting to exfiltrate data or communicate with external command-and-control servers during the evaluation pose a safety risk. The authors should confirm that the Docker containers used for evaluation have network access disabled or are strictly firewalled to prevent unintended external communication.

Finally, the paper cites Gebru et al. (Datasheets for Datasets) in the critical elements list but does not appear to include a formal datasheet or a detailed "Ethics Statement" in the main body or appendix that covers data provenance, potential biases in the selected GitHub issues (e.g., over-representation of certain languages or project types), and the limitations of the safety guarantees. Adding a dedicated subsection on "Safety and Ethical Considerations" that addresses network isolation, vulnerability sanitization, and license compliance would strengthen the manuscript's adherence to safety standards.

Action items.

- **[writing]** The 'License and Ethics' section (Appendix) acknowledges dual-use risks (autonomous exploitation) but lacks a concrete mitigation strategy beyond releasing instance IDs. Authors should explicitly state if they have redacted specific vulnerability details from the GitHub issues or if the benchmark intentionally includes known CVEs, and provide a responsible disclosure protocol for users who discover new exploits via the agents.
- **[writing]** The benchmark relies on real GitHub repositories with varying licenses (BSD, Apache, GPL-2). The paper mentions a 'REPO_LICENSES.md' file but does not detail the mechanism for ensuring downstream users comply with these licenses when running agents that might generate derivative code. A clearer statement on the legal boundaries of the generated patches and the user's responsibility regarding license compatibility is required.
- **[writing]** The evaluation involves running untrusted code (agent-generated patches) in Docker containers. While the paper mentions 'future-commit cleanup' to prevent leakage, it does not explicitly address the risk of the agents executing malicious payloads (e.g., data exfiltration, network calls to external IPs) during the 3600s timeout. A statement confirming that network access is disabled or strictly monitored in the execution environment is necessary for safety.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a well-structured benchmark for isolating the agent harness variable in coding tasks. The experimental design, specifically the "Claw-SWE-Bench" protocol with a standardized adapter and orchestrator, is a strong methodological contribution. The workload composition (350 instances, 8 languages) is substantial and appropriate for the claims made.

However, the scientific evidence supporting the quantitative conclusions is weakened by the lack of statistical replication. The "Experimental Setup" (Section 5) and "Appendix D" explicitly state "Repeats per instance: 1". All reported Pass@1 metrics (e.g., 73.4% vs 71.1% in Table 2) are single-run aggregates. In stochastic LLM-based systems, a single run per instance introduces high variance. Consequently:

1. **Statistical Significance:** The claim that "harness choice is a first-order factor" (Abstract) relies on point estimates that may not be statistically distinguishable from noise without variance estimates (standard error or confidence intervals). The 0.4pp difference between Lite and Full (Section 4) is particularly vulnerable to this, as it is likely within the margin of error for a single-run experiment.
2. **Effect Size Robustness:** The reported spreads (12.5pp and 27.4pp) are descriptive

statistics of a single realization. Without multi-seed replication (e.g., running each instance 3-5 times with different random seeds), it is impossible to determine if the observed ranking of harnesses is stable or an artifact of specific random seeds.

3. **Confounding Variables:** In Appendix D.1, the "openclaw" harness lists a default model of "claude-opus-4.6". The results in Table 2 claim to use GLM 5.1 and Qwen 3.6-flash. The text must explicitly confirm that the "openclaw" rows in Table 2 were forced to use these specific models and did not inadvertently use the harness default, which would confound the "harness" variable with a "model" variable.

The paper should be revised to either include multi-seed replication results to provide error bars or to significantly temper the language regarding the stability and statistical significance of the observed differences.

Action items.

- [**science**] Report variance estimates (standard deviation or confidence intervals) for Pass@1 metrics. The paper relies on single-run aggregates (n=1 per instance) for all 350 tasks. Without variance estimates, small differences (e.g., the 0.4pp gap between Lite and Full) cannot be statistically distinguished from noise, undermining claims of 'parity' and 'first-order' effects.
- [**science**] Clarify the statistical significance of the harness effect sizes. The paper claims harness choice changes Pass@1 by 12.5pp and 27.4pp. Given the single-run design, these are point estimates. The authors should either perform a multi-seed replication (e.g., 3-5 seeds) to establish stability or explicitly frame these as 'observed point estimates' rather than robust effect sizes in the abstract and conclusion.
- [**science**] Address the potential confounding of 'harness' with 'default model' in the claw sweep. While the paper controls the model for the 5-claw x 2-model sweep, the 'openclaw' harness is described as having a default model of 'claude-opus-4.6' (Appendix D.1), whereas others use GLM/Qwen. Ensure the reported 'openclaw' results in Table 2 strictly used the forced GLM/Qwen backbones and not the harness defaults, or clarify if the harness effect includes model drift.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The manuscript presents a comprehensive benchmark for evaluating agent harnesses, but the statistical analysis of the results requires strengthening to support the quantitative claims made.

1. Absence of Variance Estimates and Significance Testing The core findings rely on comparing Pass@1 rates across different models and harnesses. For instance, Section 5.4

states that "harness choice changes Pass@1 by 12.5 pp (GLM 5.1) and 27.4 pp (Qwen 3.6-flash)." These are point estimates derived from a single run per instance (N=350). Without reporting confidence intervals (CIs) or standard errors, it is impossible to determine if these differences are statistically significant or if they could arise from random variance in the task difficulty distribution. Given the binary nature of the outcome (resolved/not resolved), the standard error for a proportion p is $\sqrt{p(1-p)/n}$. For $p \approx 0.7$ and $n = 350$, the standard error is roughly 2.4%. A difference of 12.5pp is likely significant, but a difference of 3-4pp (as seen in the Lite validation) is not. The authors should calculate and report 95% CIs for all Pass@1 metrics in Tables 1, 2, and 3.

2. Single-Run Limitations The Discussion (Section 7) correctly identifies that "single-run aggregates" are a limitation and that "differences of only a few percentage points should not be overinterpreted." However, the Results section presents these single-run aggregates as definitive facts without qualifying the uncertainty. The statistical analysis section (or the Results introduction) must explicitly state that the reported metrics are point estimates with unknown variance due to the lack of multiple seeds. If the authors cannot re-run experiments with multiple seeds, they must frame the results as "observed differences in this specific run" rather than generalizable performance gaps, or perform a bootstrap analysis on the instance-level outcomes to estimate the variance of the aggregate metric.

3. Statistical Justification for Lite-80 Subset Section 4.2 describes the selection of the Lite-80 subset using an optimization objective (minimizing L1 difference, ranking hinge, etc.). While the resulting mean difference is small (0.4 pp), the paper lacks a formal statistical test to validate the claim that the subset is representative. A simple mean difference does not prove equivalence. The authors should perform an equivalence test (e.g., Two One-Sided Tests, TOST) or report the 95% CI of the difference between the full and lite sets to demonstrate that the difference falls within a pre-defined equivalence margin (e.g., ±2%). Currently, the claim that Lite-80 "approximates full-350 behavior" is supported only by point estimates, not statistical evidence.

4. Multiple Comparisons The paper reports results for 9 models and 5 harnesses across multiple metrics (Pass@1, Cost, Duration). While the primary focus is on Pass@1, the sheer number of comparisons increases the risk of Type I errors if any hypothesis testing were performed. Since the authors are currently reporting descriptive statistics, this is less critical, but if they proceed to claim "Model X is significantly better than Model Y," they must apply a correction for multiple comparisons (e.g., Bonferroni or Benjamini-Hochberg) to the p-values.

In summary, the experimental design is sound, but the statistical reporting is insufficient for a benchmark paper intended to establish new standards. The addition of confidence intervals and a more rigorous discussion of the single-run variance is required.

Action items.

- [**science**] Report confidence intervals or standard errors for Pass@1 metrics. The paper claims harness choice changes Pass@1 by 12.5pp and 27.4pp (Section 5.4), but without variance estimates (e.g., from bootstrapping or multiple seeds), the statistical significance of these differences cannot be assessed.

- **[science]** Address the lack of multiple-seed replication. The authors acknowledge in the Discussion (Section 7) that single-run aggregates limit the stability of small differences. The statistical analysis section must explicitly state that p-values or significance tests are not applicable due to N=1 per configuration, or provide a plan for variance estimation.
- **[science]** Clarify the statistical basis for the Lite-80 subset selection. Section 4.2 describes an optimization loss (L1, ranking hinge) but does not provide a statistical power analysis or a formal test (e.g., equivalence testing) to justify that the 0.4pp difference is statistically indistinguishable from zero rather than just numerically small.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript demonstrates a generally high standard of technical writing, with clear definitions of the benchmark, adapter protocol, and experimental setup. The logical flow from the problem statement (conflation of harness and model) to the proposed solution (Claw-SWE-Bench) is coherent. However, several areas require polishing to meet the highest standards of academic prose.

First, there are minor grammatical and syntactic issues that disrupt the reading flow. In the Introduction, the sentence "A K-sweep shows K* [8,10]; we release K=10" is abrupt and lacks a clear subject-verb connection for the second clause. It should be rephrased to explicitly state the authors' action, e.g., "We release K=10, selected from a K-sweep indicating K* [8,10]." Similarly, in Section 3.2, the enumeration of adapter methods lacks consistent punctuation and a final period, which should be corrected for professional presentation.

Second, while the data presentation is robust, some explanatory text could be more precise. In Section 5.3, the statement regarding Pass@1 changes ("12.5 pp" and "27.4 pp") is technically accurate but could be misinterpreted as a single value rather than a range/spread. Explicitly labeling these as "spreads" or "ranges" would eliminate ambiguity.

Finally, the appendices contain formatting inconsistencies that affect readability. Specifically, in Appendix D.1, the table listing denied tools for the `openclaw` harness has a row where the tool names are split across lines in a way that breaks the column alignment. This should be fixed using LaTeX's `multirow` or by adjusting the column width to ensure the table renders cleanly.

Overall, the paper is well-structured and the writing is clear, but these specific edits will enhance the professional quality and readability of the final manuscript.

Action items.

- **[writing]** In Section 1 (Introduction), the sentence 'A K-sweep shows K* [8,10]; we release K=10' is grammatically incomplete and lacks a subject for the verb 'release'. Rephrase to 'We release K=10 based on a K-sweep showing K* [8,10]' for clarity.

- [**writing**] In Section 3.2 (Adapter Protocol), the list of methods 'create_agent, send_task...' uses inconsistent spacing around commas and lacks a concluding period. Standardize the list punctuation and ensure the sentence ends with a period.
- [**writing**] In Section 5.3 (Variation Along the Claw Axis), the phrase 'Harness choice changes Pass@1 by 12.5 pp (GLM 5.1) and 27.4 pp (Qwen 3.6-flash)' is slightly ambiguous regarding which harness corresponds to which value. Clarify that these are the *spreads* (max minus min) observed for each model.
- [**writing**] In Appendix D.1 (openclaw), the table row 'Denied & sessions_list, sessions_history, & Disabled' is split across lines awkwardly, breaking the table alignment and readability. Merge the tool names into a single cell or use a multirow environment.